

KAI-HSIN HUNG

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EDUCATION

UNIVERSITY OF THE PACIFIC, Stockton, California <i>Master of Science in Data Science, GPA: 3.96 / 4.00</i> <i>Relevant Coursework: Machine Learning, Natural Language Processing, Relational Databases</i>	May 2026
MING CHI UNIVERSITY OF TECHNOLOGY, New Taipei City, Taiwan <i>Bachelor of Science in Materials Science and Engineering</i>	Jun. 2023

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, Swift
ML & AI: PyTorch, TensorFlow/Keras, YOLOv8, Scikit-Learn, spaCy, Google Gemini API
Data & Visualization: Pandas, NumPy, Plotly, Matplotlib, PostgreSQL, PostGIS
Tools & Infrastructure: Docker, Google Cloud (Cloud Run, Firebase), FastAPI, Flask, React, Git, GitHub Actions (CI/CD)

EXPERIENCE

BirthdayMessaging.io, London, UK (Remote) <i>Software Engineer Intern</i> - Synchronized 1,000+ unstructured contact records into a production database with zero manual entry, building a Python ingestion pipeline with a custom .VCF parser that parsed and validated inconsistent contact data. - Automated 100% of daily email dispatch by engineering a cron-based task scheduler to query time-sensitive events, eliminating manual intervention. - Maintained 100% data integrity across message deliveries by designing idempotent state-management logic that prevented duplicate concurrent updates.	Mar. 2025 – May 2025
Formosa Plastics Group, Taiwan <i>Manufacturing Engineer Intern</i> Organized and cleaned daily plant operational reports submitted by multiple teams, restructuring inconsistent, manually-entered records into clean, analytically-ready datasets that supported routine reporting.	Nov. 2021 – Sep. 2022

PUBLICATION

K. Hung, G. Phung, F. Lorenzino, "GoPark: An AI-Powered Parking Recommendation System," *Intelligent Systems Conference (IntelliSys) 2026*, Springer Lecture Notes in Networks and Systems. (Accepted)

SELECTED PROJECTS

AI-Driven Toxicity Prediction & Data Curation (Capstone) <i>Python, Scikit-Learn, Gemini API, Pandas, REST APIs, NLP</i> - Built the linear baseline (0.42 mean per-target R^2 across 31 endpoints) that calibrated a 5-person, 7-model toxicology benchmark and motivated the non-linear models reaching 0.58 R^2, engineering Morgan-fingerprint and descriptor features from SMILES, dose, and duration on 30,106 records. - Harmonized the DrugMatrix toxicogenomics database into a modeling-ready schema feeding the team's 30,106-record training set, integrating differential expression, clinical chemistry, and histopathology with PubChem SMILES/CASRN enrichment. - Curated 2 of 10 hepatotoxicity sources (ATSDR, CEBS) in a team standardization effort, reconciling CAS and InChIKey identifiers into unified, de-duplicated records for modeling. - Replaced manual abstract review by automating drug-hepatotoxicity literature mining over PubMed with the Gemini LLM, producing structured, confidence-scored extractions.	May 2026
GoPark: AI-Powered Parking Recommendation System (Accepted, IntelliSys 2026) <i>Python, YOLOv8, PyTorch, FastAPI, PostgreSQL/PostGIS, spaCy, SwiftUI</i> - Achieved 97.2% mAP@0.5 at 5.4 ms inference latency on edge hardware by training and deploying a YOLOv8-Nano detection model, enabling real-time parking occupancy monitoring. - Increased vehicle detection recall 5.3x under low-light conditions with a Night-Time Optimization strategy using HSV augmentation. - Built a rule-based NLP parser with spaCy, achieving 96% constraint extraction accuracy at <2 ms latency (~500x faster than GPT-4) for interpreting natural language parking queries.	Dec. 2025
JobFit AI: Intelligent Resume Analysis System <i>Python, Flask, Gemini API, PyMuPDF, React, Docker, Google Cloud Run</i> - Boosted user retention 30% with an AI resume analyzer that maps candidate skills to live job postings and surfaces gaps, built on the Gemini API and Adzuna feeds with fuzzy skill-matching. - Developed a PDF parsing pipeline with PyMuPDF to extract structured text from unstructured resumes, normalizing diverse formatting styles into a standardized skill taxonomy for LLM analysis. - Architected a full-stack streaming solution using Server-Sent Events (SSE) on Cloud Run and Firebase, reducing perceived response latency by 40%.	Jul. 2025